Stack ADT, Operations & Array Implementation

Q. Define Stack as an Abstract Data Type (ADT). Explain its primary operations and demonstrate how a Stack can be implemented using an Array with suitable code snippets.

> **Definition to Remember**

> A **S
remov
empty

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1. Stack as an ADT

| Operation | Description | Condition |

| :--- | :

2. Stack Diagram

```

Initial (

30  top

20

```

3. Array Implementation in C

```c

#includ

int stack[MAX];

int top

void push(int value) {

if (top

int pop() {

if (top
```

int peek() {

if (top

int isEmpty() { return top == -1; }

```

### 4. Advantages and Limitations

| | Array Implementation |

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> **Must-Write Points (for 10 marks)**

> 1. St

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> **Quick Recall**

> `Stac

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## Complete Executable C Code: Array Implementation of Stack ADT

```c

#includ

#define MAX 5

struct Stack {

int iten

void initStack(struct Stack *s) {

s->top

```
int isFull(struct Stack *s) {
    return
int isEmpty(struct Stack *s) {
    return
void push(struct Stack *s, int val) {
    if (isFu
int pop(struct Stack *s) {
    if (isEm
int peek(struct Stack *s) {
    if (isEm
void display(struct Stack *s) {
    if (isEm
int main() {
    struct
push(&s, 10);
    push(&
printf("Popped Value: %d\n", pop(&s));
    displa
printf("Top Element (Peek): %d\n", peek(&s));
```

return